

qtgetblk Function Test Results

Test Case 1: Single 4x4 Block

Input Matrix:

16. 2. 3. 13.

5. 11. 10. 8.

9. 7. 6. 12.

4. 14. 15. 1.

Decomposition Matrix:

4. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

Blocks:

16. 2. 3. 13.

5. 11. 10. 8.

9. 7. 6. 12.

4. 14. 15. 1.

Rows:

1.

Cols:

1.

Test Case 2: Four 2x2 Quadrants

Input Matrix:

1. 2. 5. 6.

3. 4. 7. 8.

9. 10. 13. 14.

11. 12. 15. 16.

Decomposition Matrix:

2. 0. 2. 0.

0. 0. 0. 0.

2. 0. 2. 0.

0. 0. 0. 0.

Block 1:

1. 2.

3. 4.

Block 2:

9. 10.

11. 12.

Block 3:

5. 6.

7. 8.

Block 4:

13. 14.

15. 16.

Rows:

1. 3. 1. 3.

Cols:

1. 1. 3. 3.

Test Case 3: Mixed Block Sizes

Input Matrix:

1. 9. 17. 25. 33. 41. 49. 57.

2. 10. 18. 26. 34. 42. 50. 58.

3. 11. 19. 27. 35. 43. 51. 59.

4. 12. 20. 28. 36. 44. 52. 60.
5. 13. 21. 29. 37. 45. 53. 61.
6. 14. 22. 30. 38. 46. 54. 62.
7. 15. 23. 31. 39. 47. 55. 63.
8. 16. 24. 32. 40. 48. 56. 64.

Decomposition Matrix:

4. 0. 0. 0. 2. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
2. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 1. 0.
0. 0. 0. 0. 0. 0. 0. 0.

Blocks:

(:, :, 1)

5. 13.
6. 14.

(:, :, 2)

33. 41.
34. 42.

Rows:

5. 1.

Cols:

1. 5.

Test Case 4: Corner 1x1 Blocks

Input Matrix:

16. 2. 3. 13.

5. 11. 10. 8.

9. 7. 6. 12.

4. 14. 15. 1.

Decomposition Matrix:

1. 0. 0. 1.

0. 0. 0. 0.

0. 0. 0. 0.

1. 0. 0. 1.

Blocks:

(:, :, 1)

16.

(:, :, 2)

4.

(:, :, 3)

13.

(:, :, 4)

1.

Rows:

1. 4. 1. 4.

Cols:

1. 1. 4. 4.

Test Case 5: Central 2x2 Block

Input Matrix:

1. 7. 13. 19. 25. 31.

2. 8. 14. 20. 26. 32.

3. 9. 15. 21. 27. 33.

4. 10. 16. 22. 28. 34.

5. 11. 17. 23. 29. 35.

6. 12. 18. 24. 30. 36.

Decomposition Matrix:

0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0.

0. 0. 2. 0. 0. 0.

0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0.

Blocks:

15. 21.

16. 22.

Rows:

3.

Cols:

3.

Test Case 6: No Matching Block Size

Input Matrix:

16. 2. 3. 13.

5. 11. 10. 8.

9. 7. 6. 12.

4. 14. 15. 1.

Decomposition Matrix:

4. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

Blocks:

[]

Rows:

[]

Cols:

[]

Test Case 7: Empty S

Input Matrix:

16. 2. 3. 13.

5. 11. 10. 8.

9. 7. 6. 12.

4. 14. 15. 1.

Decomposition Matrix:

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

Blocks:

[]

Rows:

[]

Cols:

[]

Test Case 8: Scattered 2x2 Blocks

Input Matrix:

1. 9. 17. 25. 33. 41. 49. 57.
2. 10. 18. 26. 34. 42. 50. 58.
3. 11. 19. 27. 35. 43. 51. 59.
4. 12. 20. 28. 36. 44. 52. 60.
5. 13. 21. 29. 37. 45. 53. 61.
6. 14. 22. 30. 38. 46. 54. 62.
7. 15. 23. 31. 39. 47. 55. 63.
8. 16. 24. 32. 40. 48. 56. 64.

Decomposition Matrix:

0. 0. 0. 0. 2. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
2. 0. 0. 0. 2. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.

Rows:

5. 1. 5.

Cols:

1. 5. 5.

Block 1:

5. 13.

6. 14.

Block 2:

33. 41.

34. 42.

Block 3:

37. 45.

38. 46.

Test Case 9: Two Decompositions

Input Matrix:

1. 5. 9. 13.

2. 6. 10. 14.

3. 7. 11. 15.

4. 8. 12. 16.

Decomposition Matrix:

2. 0. 0. 0.

0. 0. 0. 0.

0. 0. 2. 0.

0. 0. 0. 0.

Blocks:

(:,:,1)

1. 5.

2. 6.

(:,:,2)

11. 15.

12. 16.

Linear Indices:

1. 11.

Test Case 10: One Decomposition

Input Matrix:

1. 5. 9. 13.

2. 6. 10. 14.

3. 7. 11. 15.

4. 8. 12. 16.

Decomposition Matrix:

2. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

Blocks:

1. 5.

2. 6.

Test Case 11: Error Case

Input Matrix:

[]

Decomposition Matrix:

[]

Output:

"qtgetblk: Wrong number of input/output arguments."